



**Crest House and Casa Loma Recreation Residences
AMC/LBP Survey
Cibola National Forest**

Crest House...701 Sandia Crest Road, Sandia Park, New Mexico, 87047
Lat/Long 35.210262, -106.449416

Dyer House...91 Casa Loma Road, Cedar Crest, New Mexico, 87008
Lat/Long 35.109864, -106.387267

Del Toro House... 69 Casa Loma Road, Cedar Crest, New Mexico, 87008
Lat/Long 35.110333, -106.384337

Presented to:

Hailey Stock COR
Assistant Regional Environmental Engineer
USDA Forest Service
Southwestern Region

&

Danae Giersch
Forest Project Inspector/Government Technical Monitor
Civil Engineer
USDA Forest Service
Cibola National Forest

Red J Environmental Corporation

UEI: D6MLVMWQD3F8

Project Number

1-25FS009

September 23, 2025

Date of On-Site Inspection: Aug. 27, 2025

Re: Asbestos and Lead-Based Paint Inspection Report

Property Addresses: Crest House...701 Sandia Crest Road, Sandia Park, New Mexico, 87047

Dyer House...91 Casa Loma Road, Cedar Crest, New Mexico, 87008

Del Toro House... 69 Casa Loma Road, Cedar Crest, New Mexico, 87008

Property Owner: Cibola National Forest

Dear Client:

Please find enclosed the asbestos and lead-based paint inspection report for the property located at the aforementioned locations. The survey was performed within the current acceptable industry guidelines, and EPA regulations. Red J Environmental Corporation (Red J) conducted the asbestos surveys and lead-based paint inspection services at the above-referenced sites on August 27, 2025. Licensed OSHA Lead Sampling Technician Vonnie Young (Certification No. 2159-ANTLS72925-8012, expiration date 07/08/2026) performed the inspection services. Licensed EPA AHERA Building Inspector (Certification No. 2604-ANTRBI81325-8012)

At the specific time and date of the inspection services, Red J detected no lead-based paint in the property. The asbestos tested did find two locations at separate sites that do contain asbestos above industry guidelines.

If you have any questions or concerns regarding this report, please feel free to contact us at (928)288-3239.

Sincerely,

Stephen Solomon

Stephen Solomon
General Manager

Table of Contents

Executive Summary.....	1
Asbestos Overview.....	1
Lead Overview.....	5
Limitations.....	5
Scope of Surveys and Inspections.....	7
Asbestos Recommendations.....	19
Lead Recommendations.....	19
Certifications/Laboratory Report/Chain of Custody	20
Glossary.....	27

Executive Summary

On August 27th, 2025, Red J Environmental Corporation (Red J) was authorized to conduct a limited Asbestos Containing Material (ACM) Survey and Lead-Based Paint (LBP) Inspection of three separate properties located within the Cibola National Forest. A Kick-Off meeting was conducted on Aug. 19th, 2025 to discuss the activities to take place at two (2) of the properties which are single-family detached dwellings and are believed to be built between 1940 and 1950. The third property is a combination of former Commercial and Residential use and was constructed in 1978.

Red J tested all components according to the specifications described in the protocols for a limited ACM survey/testing and LBP inspection/testing, within all applicable Federal, State, and Local regulations. This Survey/Inspection was conducted on site by an EPA-trained professional asbestos building inspector.

This report contains the results of the Surveys and Inspections conducted. This report is intended for the exclusive use of our client. The findings are relevant to the conditions observed during the physical process of performing the Survey. These findings should not be treated as absolute, nor should they be relied upon to represent conditions at significantly later dates.

Asbestos Overview

The purpose of this ACM Survey was to identify the presence of asbestos-containing materials that may be disturbed during a planned demolition. Limited bulk samples were collected and AHERA protocols were adhered to. RED J tested a total of eleven (11) surfaces. Two (2) were found to contain asbestos at levels greater than or equal to the EPA and NESHAP recommendations using the point-counting procedure for confirmation of asbestos percentage in friable materials that are visually estimated by Polarized Light Microscopy (PLM) methodology to contain less than 10% asbestos. The 400 Point Count Procedure referenced in EPA 600/M4-82-020 (1987) and EPA 600/R-93/116 (1993) is commonly employed. Without the material being point counted or if point counting determined that material contains greater than one percent asbestos, it would be deemed an asbestos containing material and would need to be removed by a New Mexico licensed asbestos contractor prior to disturbance.

The Asbestos Survey consisted of three basic procedures:

- 1) conducting a visual inspection of the property
- 2) identifying homogeneous areas (HAs) of suspect surfacing, thermal system insulation, and miscellaneous materials
- 3) sampling accessible, friable, and non-friable suspect materials.

Some building components may have been inaccessible at the time of this screening, or were not tested because they were covered by other building materials (paneling, tile, siding, etc.). It is possible that ACMs may be hidden by these materials. The property was visually inspected for the presence of building materials that are suspected to contain asbestos. With regard to asbestos, bulk material samples were collected and analyzed for asbestos content. These services were performed exercising the customary skill and competence of consulting professionals in the relevant disciplines in this region. Bulk samples of identified suspect ACM were collected and placed into individual containers for transport to A.E.S.L Environmental Laboratory, a National Voluntary Lab Accreditation Program (NVLAP)/American Industrial Hygiene Association (AIHA)-accredited laboratory for analysis. The collection of bulk samples consisted of physically removing a small piece of material and placing it in a marked, airtight container. The sample container identification numbers were also recorded in the field notes.

Asbestos is a generic name given to a fibrous variety of naturally occurring minerals that have been used for many years in commercial products, based on specific properties of the minerals. Asbestos occurs in fiber bundles, which are composed of long and thin fibers that can be easily separated from one another. These mineral products possess high tensile strength, flexibility, resistance to chemical and thermal degradation, and high electrical resistance. The minerals are easily woven into various types of textiles, fabrics, cloths, sheets, panels, or mixed into adhesives, coatings, surfacing materials and cement products.

Typically, asbestos-containing building materials (ACBM) are segregated into three categories:

- Thermal System Insulation (TSI) usually found on pipes, boilers, and HVAC ducts; surfacing materials such as sprayed or troweled-on fireproofing and insulation, and plasters; and miscellaneous materials including vinyl composite floor tiles, floor sheeting, adhesives, roofing materials, window glazing and cement products.
- *Friable* asbestos-containing material (ACM), is defined as any material containing more than one percent (1%) asbestos as determined using the method specified in Appendix A, Subpart F, 40 CFR Part 763, Section 1, Polarized Light Microscopy (PLM), that, when dry, can be crumbled, pulverized or reduced to powder by hand pressure. (Sec. 61.141)
- *Nonfriable* ACM is any material containing more than one percent (1%) asbestos as determined using the method specified in Appendix A, Subpart F, 40 CFR Part 763, Section 1, Polarized Light Microscopy (PLM), that, when dry, cannot be crumbled, pulverized, or reduced to powder by hand pressure. EPA also defines two categories of nonfriable ACM, Category I and Category II nonfriable ACM, which are described later in this guidance.

"Regulated Asbestos-Containing Material" (RACM) is

- (a) friable asbestos material
- (b) Category I nonfriable ACM that has become friable
- (c) Category I nonfriable ACM that will be or has been subjected to sanding, grinding, cutting or Abrading
- (d) Category II nonfriable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of demolition or renovation operations.

The EPA's National Emission Standard for Hazardous Air Pollutants (NESHAP) regulations regulate the removal and disposal of asbestos-containing building materials (any material containing more than 1% asbestos). Potential effects on workers or occupants in buildings where asbestos-containing materials (ACM) are present may occur when exposure to asbestos fibers is caused by deterioration, damage or renovation disturbance of ACBMs. Federal regulations pertaining to asbestos include 40 Code of Federal Regulations (CFR) 763 (a subchapter of the Toxic Substance Control Act (TSCA)); Occupational Safety and Health Act (OSHA) 29 CFR 1910 Subpart Z and 29 CFR 1926 Subpart Z.

Asbestos NESHAP regulations must be followed for demolitions and/or renovations of facilities with at least 260 linear feet of regulated asbestos containing materials (RACM) on pipes, 160 square feet of regulated asbestos-containing materials on other facility components, or at least 35 cubic feet of facility components where the amount of RACM previously removed from pipes and other facility components could not be measured before stripping. If dimensions fall below these thresholds, Asbestos NESHAP regulations need not be followed for demolition and/or renovation activities.

Background on the Asbestos NESHAP

EPA's air toxics regulation for asbestos is intended to minimize the release of asbestos fibers during activities involving the handling of asbestos. The air toxics provisions of the Clean Air Act (CAA) require EPA to develop and enforce regulations to protect the public from exposure to airborne contaminants that are known to be hazardous to human health. In accordance with Section 112 of the CAA, EPA establishes National Emission Standards for Hazardous Air Pollutants (NESHAP). The list of hazardous air pollutants (HAP), or "air toxics", includes specific compounds that are known or suspected to cause cancer or other serious health effects. Three of the major health effects associated with asbestos exposure are lung cancer, mesothelioma, and asbestosis. Asbestos was one of the first hazardous air pollutants regulated under the air toxics program. On March 31, 1971, EPA identified asbestos as a hazardous pollutant, and on April 6, 1973, EPA promulgated the Asbestos NESHAP, currently found in 40 CFR Part 61, Subpart M. The Asbestos NESHAP has been amended several times, most comprehensively in November 1990. In 1995, the rule was amended to correct cross-reference citations to OSHA, DOT, and other EPA rules governing asbestos.

Renovation and Demolition of Buildings

Air toxics regulations under the Clean Air Act specify work practices for asbestos to be followed during demolitions and renovations of all facilities, including, but not limited to, structures, installations, and buildings (excluding residential buildings that have four or fewer dwelling units). The regulations require a thorough inspection where the demolition or renovation operation will occur. The regulations require the owner or the operator of the renovation or demolition operation to notify the appropriate delegated entity (often a state agency) before any demolition, or before any renovations of buildings that contain a certain threshold amount of regulated asbestos-containing material. The rule requires work practice standards that control asbestos emissions. Work practices often involve removing all asbestos-containing materials, adequately wetting all regulated asbestos-containing materials, sealing the material in leak tight containers and disposing of the asbestos-containing waste material as expediently as practicable, as the regulation explains in greater detail. These work practice standards are designed to minimize the release of asbestos fibers during building demolition or renovation, waste packaging, transportation and disposal.

Institutional, Commercial and Industrial Buildings

Any demolition or renovation operation at an institutional, commercial or industrial building is regulated by the Asbestos NESHAP. At a minimum, the thorough inspection requirement applies. The notification requirements apply to any demolition and to renovations over a certain threshold amount of regulated asbestos-containing material.

Large-Scale Residential Demolition

In 1990, EPA amended the NESHAP for asbestos. In the revised rule, EPA noted that it does not consider residential structures that are demolished or renovated as part of a commercial or public project to be exempt from the NESHAP. For example, EPA explained that the demolition of one or more houses as part of an urban renewal project, a highway project, or a project to develop a shopping mall, industrial facility, or other private development would be subject to the NESHAP. In 1995, EPA issued a Clarification of Intent (PDF) (2pp, 147k) regarding demolition of isolated single-family homes.

Exceptions to the Asbestos Removal Requirements

If the total amount of asbestos to be removed or disturbed is less than 260 linear feet, 160 square feet, or 35 cubic feet off of facility components (if the material could not be measured previously), then the Asbestos NESHAP does not require regulated asbestos-containing material to be removed before demolition or renovation activities occur.

Asbestos Control Method

The Asbestos NESHAP requires specific work practices to control the release of asbestos fibers. To help ensure that the work practice standards of the Asbestos NESHAP are followed during a demolition or renovation operation, the asbestos NESHAP requires at least one onsite representative trained in the regulatory provisions and the means of compliance. This trained individual needs to receive refresher training every two years, including: applicability of the rule; notifications; material identification; control procedures for removal; adequate wetting; local exhaust ventilation; negative pressure enclosures; glove-bag procedures; High Efficiency Particulate Air (HEPA) filters; waste disposal work practices; reporting and record keeping; and, asbestos hazards and worker protection. EPA has not approved any alternatives to the demolition requirements of the Asbestos NESHAP.

Waste Disposal and Transportation

The rule generally requires that asbestos-containing waste material be sealed in a leak-tight container while wet, labeled, and disposed of properly in a landfill qualified to receive asbestos waste. Landfills have special requirements for handling and securing the asbestos containing waste to prevent releases of asbestos into the air. Transportation vehicles that move the waste from the point of generation to the asbestos landfill have special labeling requirements and waste shipment record keeping requirements.

Lead Overview

Red J's scope of services for LBP involved a surface-by-surface visual inspection of all painted surfaces throughout the entire property to determine which LBP surfaces/components are deteriorated (above de minimis level).

RED J tested a total of eleven (11) surfaces. Zero (0) were found to contain lead at levels greater than or equal to the regulatory level of 1.0 mg/cm². These surfaces are identified in Section III: G. This report represents all field data, observations and findings related to the lead inspection performed in the above referenced property. The results, assessments and findings stated in this report are representative of the conditions observed in this property at the time of the inspection services.

This inspection measures lead in both deteriorated and intact paint surfaces. The procedure involved taking samples from representative surfaces throughout the testing area or room. Some building components may have been inaccessible at the time of the inspection services, or were not tested because they were covered by other building materials (paneling, tile, siding, etc.). It is possible that painted surfaces may be hidden by these materials. Such surfaces should be assumed to contain lead-based paint, or should be tested by a licensed lead-based paint inspector or risk assessor.

Limitations

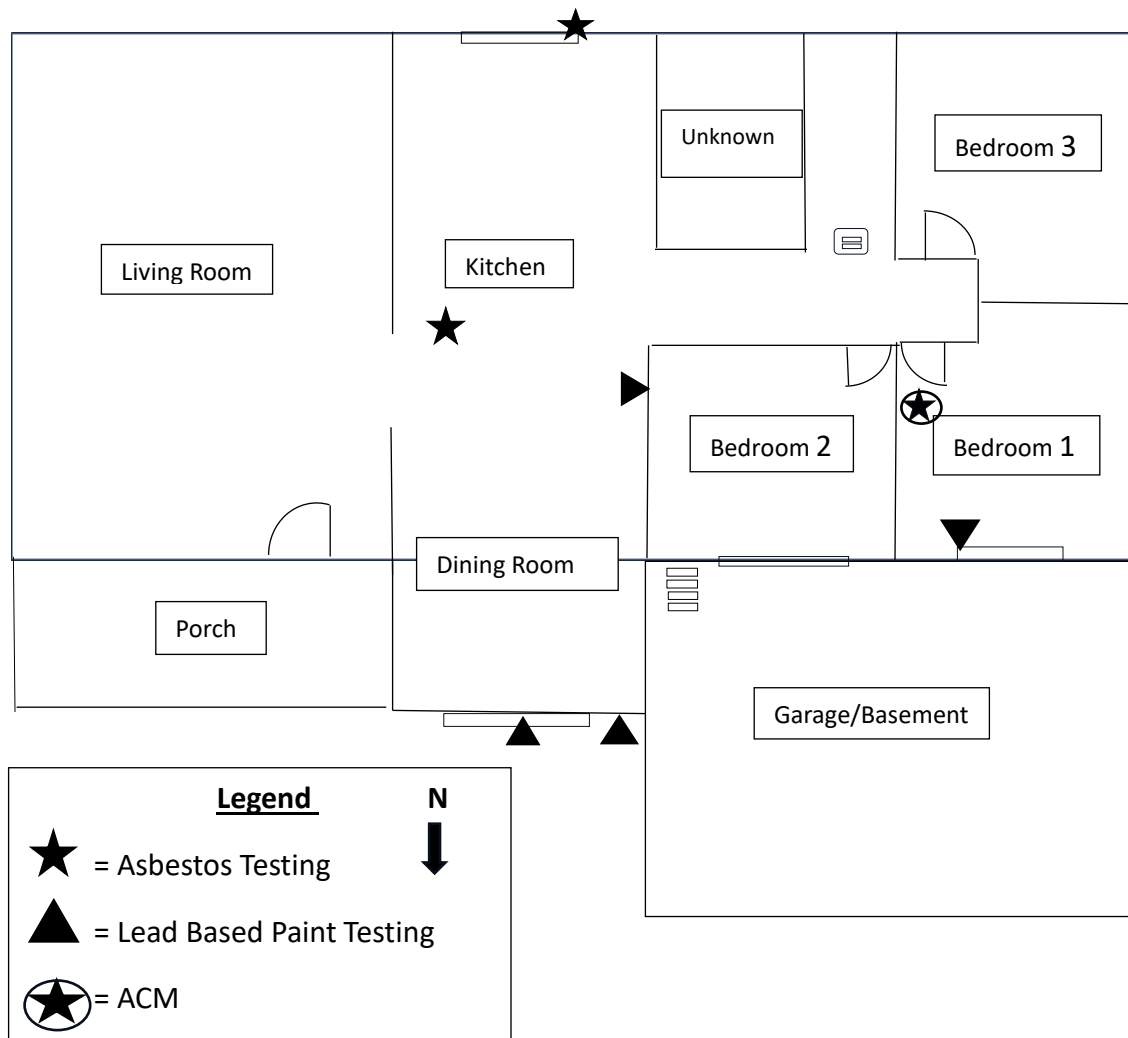
This report has been prepared to assist in evaluating the potential presence of ACM and LBP on the properties listed. The objective of this assessment was to perform the work with care, exercising the customary skill and competence of consulting professionals in the relevant disciplines in this region. The conclusions presented in this report are professional opinions based upon visual observations of the site at the time of Red J.s investigation and the results of laboratory analysis. The opinions presented herein apply to site conditions existing at the time of our investigation and those reasonably foreseeable. The results and opinions in this report,

based solely upon the conditions found on the property as of the date of the Survey, will be valid only as of the date of the Survey.

Please note that the test results relate only to those homogeneous materials tested. If conditions or materials, other than those addressed in this report are encountered during the planned renovation/demolition activities, Red J should be contacted to assess the potential impact of these materials or conditions relative to the findings or recommendations included herein. The survey was performed by observing suspect materials throughout the structure where accessible. Red J must emphasize that it is not possible to look within every location of a building. The visual survey documents only general locations of suspect materials but does not determine exact boundaries. Concealed locations of asbestos or lead paint may exist at the subject property, and the levels may vary from those stated in this report. There may be variations in the composition of materials which appear similar. Materials may be hidden from view and not accessible. No attempt was made to disassemble equipment or demolish structural elements and finishes as this is beyond the scope of our authorized services. Visual observations were made only at safe and convenient locations. Due to these limitations, wall voids, flooring under carpet, building cavities and mechanical equipment, and other areas may contain unreported asbestos-containing materials. Suspect materials not previously identified in this report may be encountered during any renovation/demolition activity. These materials should be assumed asbestos containing material or lead based paint until sample collection and subsequent analysis prove otherwise. Unsafe structures should be assumed to contain asbestos materials unless the suspect material is noted as sampled. All fire doors should be assumed asbestos containing material since disassembly of locks and/or other work to access the door insulation is not possible.

Scope of Surveys and Inspections

DYER RESIDENTIAL RESIDENCE



**Below is a photographic journal of structures and testing locations during the Dyer ACM
Surveys and LBP Inspections.**



North facing side of the Residency



East facing side of the Residency



South facing side of the Residency



West facing side of the Residency

ACM Testing Locations and Results at Dyer



Bedroom 1 Ceiling
Sample #22



Kitchen Floor
Sample #23



Roof Tile
Sample #24

<i>Bulk Collection and Sample Analysis Results</i>						
<i>Sample Number</i>	<i>Description</i>	<i>Condition</i>	<i>Friable</i>	<i>Asbestos Percent and Type</i>	<i>Location</i>	<i>NESHAP Category</i>
22	Popcorn Texture	Intact	Yes	2% Chrysotile	Bedroom 1 Ceiling	NA
23	Floor Tile	Intact	No	None Detected	Kitchen Floor	NA
24	Asphalt Shingle/Black	Intact	No	None Detected	Typical Exterior Roof	NA

Of the three samples taken at the Dyer House one of them was positive for 2% Chrysotile asbestos. That sample was #22 taken from Bedroom 1 on the ceiling. The other two samples did not detect any asbestos.

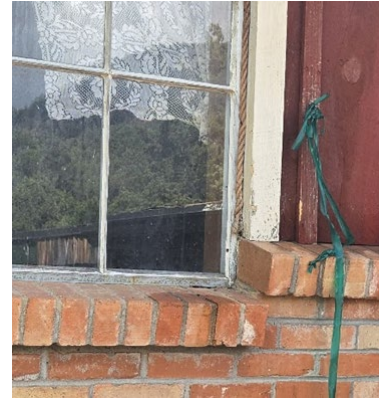
LBP Testing Locations and Results at Dyer



Bedroom 1 Window Sill
Sample #1



Diningroom Wall
Sample #2

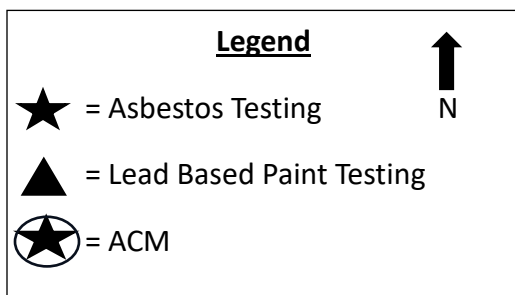
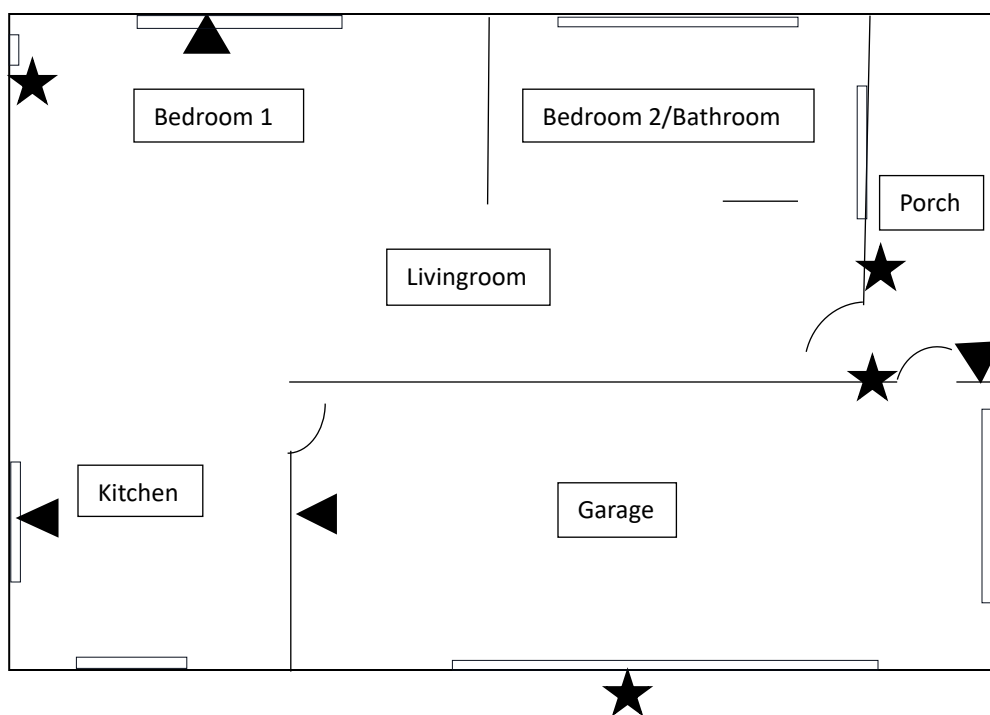


Porch Window Sill and Wall
Sample #3 & #4

<i>XRF Shot Number</i>	<i>Client Number</i>	<i>Description</i>	<i>Condition</i>	<i>Location</i>	<i>Concentration</i>
1	1	Yellow Paint	Peeling	Bedroom 1	<17ppm
2	2	Multicolor Paint	Intact	Dining Room	<17ppm
3	3	White Paint	Peeling	Porch Window	<17ppm
4	4	Maroon Paint	Peeling	Outside Wall	<17ppm

Zero (0) were found to contain lead at levels greater than or equal to the regulatory level of 1.0 mg/cm² or 5,000ppm.

DEL TORO RESIDENTIAL RESIDENCE



Below is a photographic journal of structures and testing locations during the Del Toro ACM Survey and LBP Inspection.



South facing side of the Residency



East facing side of the Residency



West facing side of the Residency



North facing side of the Residency

ACM Testing Locations and Results at Del Toro



Sample #25
Bedroom 1 wall/ceiling tile



Sample #26
Tar paper



Sample #27
Outside Shingle



Sample #28
Green Shingle

Bulk Collection and Sample Analysis Results						
Sample Number	Description	Condition	Friable	Asbestos Percent and Type	Location	NESHAP Category
25	Wall/Ceiling Tile	Intact	No	None Detected	Bedroom 1 Wall	NA
26	Tar Paper	Torn	No	None Detected	Above front door	NA
27	Shingle	Torn	No	None Detected	Above door to garage	NA
28	Green Shingle	Torn	No	None Detected	South side wall	NA

Of the four (4) samples taken at the Del Toro House none of them detected any asbestos.

LBP Testing Locations and Results at Del Toro



Sample #5
Bedroom 1 Window Sill



Sample #6
Kitchen Window Sill



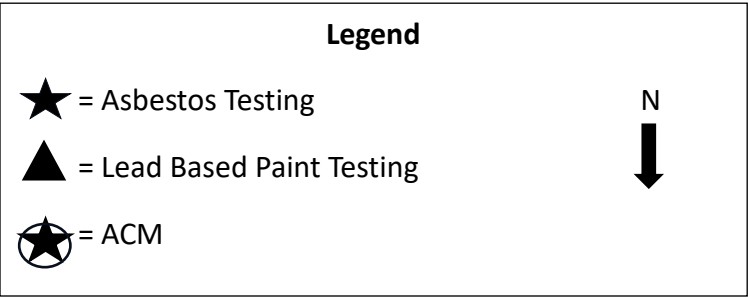
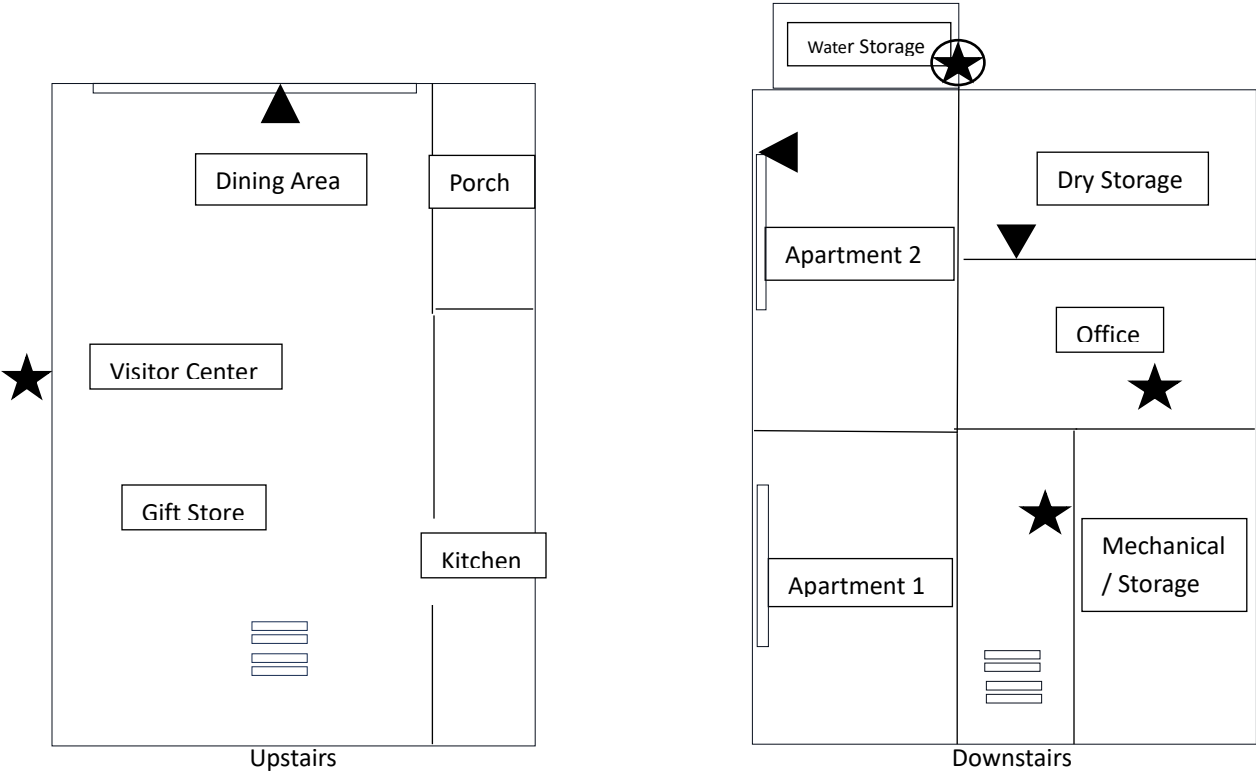
Sample #7
Garage Wall



Sample #8
Outside Wall on Porch

<i>XRF Shot Number</i>	<i>Client Number</i>	<i>Description</i>	<i>Condition</i>	<i>Location</i>	<i>Concentration</i>
5	5	Multicolor Paint	Peeling	Bedroom 1	<17ppm
6	6	Multicolor Paint	Intact	Kitchen Window	<17ppm
7	7	Blue/White Paint	Peeling	Garage Wall	<17ppm
8	8	Brown Paint	Peeling	Outside Wall	<17ppm

CREST HOUSE



**Below is a photographic journal of structures and testing locations during the Crest House
ACM Survey and LBP Inspection.**



North Side of Building



West Side of Building



South Side of Building

ACM Testing Locations and Results at Crest House



Sample # 29
Basement Storage Ceiling



Sample #30
Basement Tile



Sample #31
Outside South Wall



Sample #32
Roof Over Water Storage

<i>Bulk Collection and Sample Analysis Results</i>						
<i>Sample Number</i>	<i>Description</i>	<i>Condition</i>	<i>Friable</i>	<i>Asbestos Percent and Type</i>	<i>Location</i>	<i>NESHAP Category</i>
29	Ceiling Texture	Intact	No	None Detected	Basement Ceiling	NA
30	Floor Tile	Cracked	No	None Detected	Basement Tile	NA
31	Stucco	Cracked	No	None Detected	Outside Wall	NA
32	Roofing Mastic	Torn	Yes	3% Chrysotile	Roof Over Water Storage	NA

LBP Testing Locations and Results at Crest House



Sample # 9
Apartment 2 Wall



Sample #10
Basement Wall



Sample #11
Dining Room Wall

<i>XRF Shot Number</i>	<i>Client Number</i>	<i>Description</i>	<i>Condition</i>	<i>Location</i>	<i>Concentration</i>
9	9	White Paint	Peeling	Apartment 2	<17ppm
10	10	White Paint	Intact	Basement Wall	<17ppm
11	11	White Paint	Intact	Dining Room Wall	<17ppm

Asbestos Recommendations

The results of the asbestos survey conducted at the Dyer House show there is 2% Chrysotile at located on the ceiling in bedroom 1. Due to Federal and State laws NESHAP does not apply in this case since it is a single-family home.

According to the New Mexico Environmental Department (NMED), A home owner who has always used the home as a single-family dwelling is not subject to the federal asbestos NESHAP. Refer to 40 CFR 60 subpart M for demolition and remediation requirements for other types of residences.

The results of the asbestos survey conducted at the Sandia Crest House showed there is 3% Chrysotile in the black mastic located over the water storage on the south side of the main building. While it is a commercial building, the area affected is believed to only be up to 152 square feet, which falls under the 160 square feet requirement for NESHAP. Therefore, it too is exempt from NESHAP.

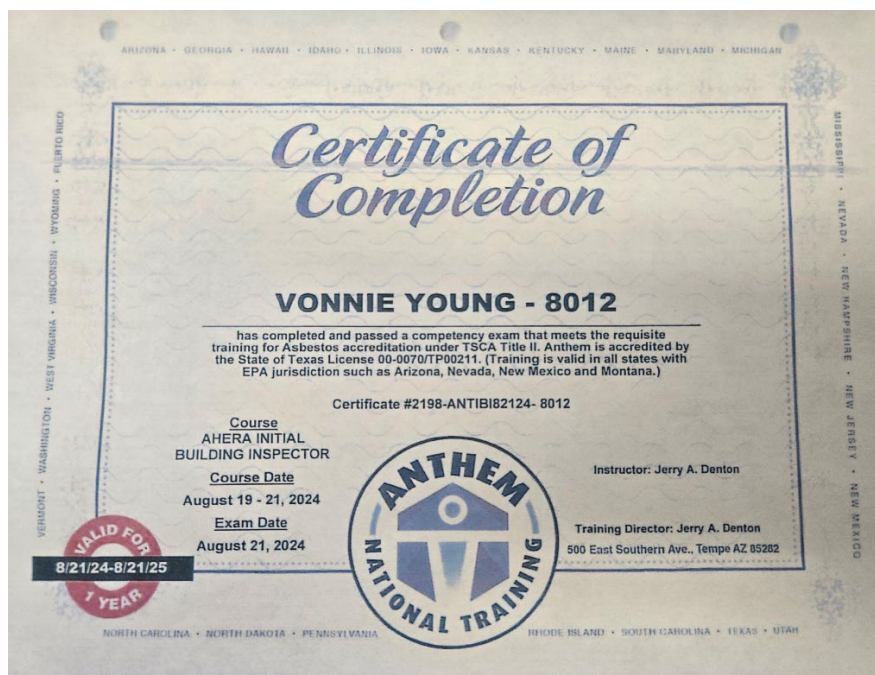
If the total amount of asbestos to be removed or disturbed is less than 260 linear feet, 160 square feet, or 35 cubic feet off of facility components (if the material could not be measured previously), then the Asbestos NESHAP does not require regulated asbestos-containing material to be removed before demolition or renovation activities occur.

Precautions should always be taken when demolition occurs where asbestos has been detected. At this time NESHAP is not required when demolition takes place at any of the aforementioned locations.

Lead Recommendations

No lead-based paint was found in any of the samples taking at the time of investigation. Precautions should always be taken during demolition of any building built before 1970.

Certifications/Laboratory Report/Chain of Custody







Anthem National Training

Vonnie Young
XXX-XX-8012

has completed a competency course that meets requirements of TSCA Title II and is certified as:

AHERA Supervisor Refresher

Class Date(s): August 15, 2025
Exam Date: August 15, 2025
Certification Exp: August 15, 2026

500 East Southern Ave., Tempe, AZ 85282
(480) 966-6500

Vonnie Young
SSN - 8012

ARIZONA • GEORGIA • HAWAII • IDAHO • ILLINOIS • IOWA • KANSAS • KENTUCKY • MAINE • MARYLAND • MICHIGAN

MISSISSIPPI • NEVADA • NEW HAMPSHIRE • NEW JERSEY • NEW MEXICO

VERMONT • WASHINGTON • WEST VIRGINIA • WISCONSIN • WYOMING • PUERTO RICO

Certificate of Completion

VONNIE YOUNG - 8012

has completed this course and meets OSHA requirements for 29 CFR 1926.62

Certificate #2159-ANTLS72925-8012

Course
OSHA LEAD SUPERVISOR

Course Date
July 29, 2025
Exam Date
July 29, 2025

INSTRUCTOR
Instructor: Jerry A. Denton

TRAINING DIRECTOR
Training Director: Jerry A. Denton
500 East Southern Ave., Tempe AZ 85282

VALID FOR 1 YEAR
7/29/25 - 7/29/26

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Certificate of Completion

VONNIE YOUNG - 8012

has completed this course and meets OSHA requirements for 29 CFR 1926.62

Certificate #2159-ANTLS72925-8012

Course
OSHA LEAD SAMPLING TECH

Course Date
July 28, 2025
Exam Date
July 28, 2025

INSTRUCTOR
Instructor: Jerry A. Denton

TRAINING DIRECTOR
Training Director: Jerry A. Denton
500 East Southern Ave., Tempe AZ 85282

VALID FOR 1 YEAR
7/28/25 - 7/28/26

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Turnaround Time: ☒ RUSH
☐ Same Day

☐ 24 Hour
☐ 48 Hour

☐ Stop @ First Positive
☐ Read All Samples

**BULK ASBESTOS SAMPLE
CHAIN OF CUSTODY**

A.E.S.L. LABORATORY #: 25-A103

Page 2 of

Client Name: <u>Red J Environmental Corp.</u>	Phone: <u>(928) 288-3239</u>	Fax: <u>(928) 288-3836</u>
Address: <u>PO Box A</u>	City: <u>Joseph City</u>	State: <u>Arizona</u> Zip: <u>86032</u>

PROJECT: Crest House and Casa Loma ACM/ABR Survey PROJECT ID: 1-25FS009

*** SAMPLES TO BE RETURNED TO CLIENT AFTER 30 DAYS OR DISPOSED OF BY A.E.S.L. (D R): ***
(If NOT SPECIFIED WILL AUTOMATICALLY BE DISPOSED OF AFTER 30 DAYS)

SAMPLE #	SAMPLE DESCRIPTION	SAMPLE LOCATION	QUANTITY
1 22	Front Bedroom Ceiling 11:16	Dyer Recreation Residence	1
2 23	Kitchen Floor Tile 11:24	Dyer Recreation Residence	1
3 24	Roof Tile 11:26	Dyer Recreation Residence	1
4 25	Side Room Wall/Ceiling Tile 11:51	Del Toro Recreation Residence	1
5 26	Outside Tar Paper 11:58	Del Toro Recreation Residence	1
6 27	Outside Shingle 11:59	Del Toro Recreation Residence	1
7 28	Outside Green Shingle 12:05	Del Toro Recreation Residence	1
8 29	Basement Storage Ceiling 12:59	Crest House	1
9 30	Basement Tile 13:01	Crest House	1
10 31	Outside Back Wall 13:09	Crest House	1
11 32	Roof over water storage 13:13	Crest House	1
33			
34			
35			
36			
37			
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40			
41			
42			

A.E.S.L. Environmental Laboratory 8500 East Southern Avenue, Suite A Tempe, Arizona 85282	RELINQUISHED BY: <u>Yannick P. Jany</u> RECEIVED BY: <u> </u>	DATE: <u>8/28/25</u> DATE: <u>8/29/25</u>
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**A. E. S. L.
Environmental
LABORATORY**

A DIVISION OF ALLEN ENVIRONMENTAL CONSULTING L.L.C.
500 East Southern Avenue, Suite A • Tempe, Arizona 85282
(480) 966-7171 • Fax (480) 394-0188

BULK ASBESTOS ANALYSIS SUMMARY REPORT

CLIENT NAME: Red J Environmental
PO Box A
Joseph City, Arizona 86032

DATE OF RECEIPT: August 28, 2025
SAMPLE CONDITION: Good
DATE ANALYZED: August 29, 2025

A.E.S.L. LABORATORY #: 25-A103

PROJECT: Crest House / Casa Loma.

A.E.S.L. LAB SAMPLE ID #	CLIENT SAMPLE ID #	SAMPLE DESCRIPTION & COLOR	TEST RESULTS		OTHER MATERIALS
			Pos. / n.d.	% & Type	
A103-1	22	Drywall Texture/Paint	Positive	2% Chrysotile	98% Non-Fibrous
A103-2	23	Tan Floor Tile/Clear Mastic	n.d.	-----	100% Non-Fibrous
A103-3	24	Brown Roof Shingle	n.d.	-----	30 % Fiberglass
A103-4	25	Wall & Ceiling Tile	n.d.	-----	100% Cellulose
A103-5	26	Outside Tar Paper	n.d.	-----	30% Cellulose
A103-6	27	Tan Roofing & Black Tar Paper	n.d.	-----	30% Cellulose
A103-7	28	Green Roofing & Black Tar Paper	n.d.	-----	25% Cellulose
A103-8	29	White Ceiling Texture	n.d.	-----	100% Non-Fibrous
A103-9	30	Beige Floor Tile & Tan Mastic	n.d.	-----	100% Non-Fibrous
A103-10	31	Grey Stucco & Tan Paint	n.d.	-----	100% Non-Fibrous
A103-11	31	Black Roof Mastic	Positive	3 % Chrysotile	97% Non-Fibrous

n.d. = Non-Detect

Method: Polarized Light Microscopy, EPA Method 600/R-93/116

The result quantitations reported are an estimation based on the methods of visual microscopic estimation, which is considered only a semi-quantitative technique. Also, this report is indicative only of the sample material A.E.S.L. Laboratory received. Results do not necessarily reflect the makeup of the entire span of the material from which the samples were derived. Sampling techniques and/or sample handling may affect the integrity of the sample/s before submission to A.E.S.L. Laboratory and hence the outcome of the laboratory results. Samples not destroyed by testing are retained a minimum of thirty days. A.E.S.L. Laboratory, recommends re-analysis by point count or Transmission Electron Microscopy (TEM) for materials that are identified to contain less than ten percent (<10%) asbestos by PLM.

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Review and QA/QC by: Jerry A. Denton, Laboratory Director

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NVLAP 200303-0

CALIF. ELAP 2345

**LEAD ANALYSIS
CHAIN OF CUSTODY**

Page 1 of 1

Client Name: <u>Red J Environmental</u>		Contact: <u>Vernice Young</u>	Phone: <u>928-288-3237</u>
Address: <u>PO Box A</u>		City: <u>Joseph City</u>	State: <u>AZ</u> Zip: <u>86032</u>

PROJECT: Crest House and Casa Loma NW/LEAD PROJECT ID: 1-25FS009 DATE SAMPLES TAKEN: 8/27/25 Survey

SAMPLES REC'D (#): _____ DATE REC'D: _____ CONDITION: _____ SAMPLES ACCEPTED (Y , N):

XRF: _____ TCLP: _____ SOIL-AA: - _____ WIPE-AA: _____ SAMPLES SUBMITTED:

Sample #	Sample Location	Sub Sample Location	Sample Description	Sample #	Sample Location	Sub Sample Location	Sample Description
1	Dyer Res. Residence	Bedroom #1 Window Frame	Paint / Yellow 1117	11	Crest House	Dining Room / Kitchen	Paint / White 1300
2	Dyer Res. Residence	Dining Room Wall	Paint / Multicolor 1121	12			
3	Dyer Res. Residence	Porch Window	Paint / White 1128	13			
4	Dyer Res. Residence	Wall Outside	Paint / Maroon 1130	14			
5	Del Toro Res. Residence	Side Room Window	Paint / Multicolor 1152	15			
6	Del Toro Res. Residence	Kitchen Window	Paint / Multicolor 1155	16			
7	Del Toro Res. Residence	Garage Wall	Paint / Blue/White 1156	17			
8	Del Toro Res. Residence	Outside Wall	Paint / Brown 1157	18			
9	Crest House	Apt #2 Wall	Paint / White 1257	19			
10	Crest House	Entryway Wall	Paint / White 1300	20			

A.E.S.L. Environmental Laboratory 800 North Mary Street, Tempe, Arizona 85281 DATE: <u>8/28/25</u> Phone (480) 966-7171 Fax (480) 393-0193		RELINQUISHED BY: <u>Vernice Young</u> <u>8/28/25</u>
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**A. E. S. L.
Environmental
LABORATORY**

A DIVISION OF ALLEN ENVIRONMENTAL CONSULTING L.L.C.
500 East Southern Avenue, Suite A • Tempe, Arizona 85282
(480) 966-7171 • Fax (480) 394-0188

LEAD ANALYSIS SUMMARY

CLIENT: Red J
PO Box A
Joseph City, Arizona 86023

PROJECT:
Crest
House/Casa
Loma

25-L034

PREPARATION METHOD: XRF

ANALYSIS METHOD: XRF 7024

XRF Shot Number	Client Number	Date Sampled	Sample Description	Concentration
1	1	8/27/2025	Yellow Paint	<17 ppm
2	2	8/27/2025	Multi Color Paint	<17 ppm
3	3	8/27/2025	White Paint	<17 ppm
4	4	8/27/2025	Maroon Paint	<17 ppm
5	5	8/27/2025	Multi Color Paint	<17 ppm
6	6	8/27/2025	Multi Color Paint	<17 ppm
7	7	8/27/2025	Blue / White Paint	<17 ppm
8	8	8/27/2025	Brown Paint	<17 ppm
9	9	8/27/2025	White Paint	<17 ppm
10	10	8/27/2025	White Paint	<17 ppm
11	11	8/27/2025	White Paint	<17 ppm

- OSHA limit is over 17 ppm7

QUALITY CONTROL DATA

BATCH #101669
Calibration Standard 1.68 mg/cm²
Calibration Verification 1.68 mg/
Laboratory Control Standard 1.68 mg/cm²
Reporting Limit 0.00 mg/cm²
Method Detection Limit 0.00 mg/cm²

Analyzed By: Jerry A. Denton

Date: 8/30/25

Results represent the analysis of samples submitted by the client. Sample location, description, area, volume, etc., was provided by the client. Although the submission of blank samples is required by sampling methodologies, sample results are not blank-corrected. This report shall not be reproduced, except in full, without the written consent of A.E.S.L Environmental Laboratory.

LEGEND ml = milliliter ug = microgram ppm = parts per million
Pb = lead L = Liters ug/m³ = micrograms per cubic meter
ug/FT² = micrograms per square foot cm² = centimeter squared
mg = milligram

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NVLAP 200303-0

CALIF. ELAP 2345

Glossary

Abatement: A measure or set of measures designed to permanently eliminate lead-based paint hazards or lead based paint. Abatement strategies include the removal of lead-based paint, enclosure, encapsulation, replacement of building components coated with lead-based paint, removal of lead-contaminated dust, and removal of lead-contaminated soil or overlaying of soil with a durable covering such as asphalt (grass and sod are considered interim control measures). All of these strategies require preparation, cleanup, waste disposal, post-abatement clearance testing, record keeping, and, if applicable, monitoring. See also **Complete Abatement** and **Interim controls**.

Accreditation: A formal recognition certifying that an organization, such as a laboratory, is competent to carry out specific tasks or types of tests.

Accuracy: The degree of agreement between an observed value and an accepted reference value (a true value); a data quality indicator. Accuracy includes a combination of random errors (Precision) and systematic errors (bias) due to sampling and analysis.

Active waste disposal site: any disposal site other than an inactive site.

Adequately wet: sufficiently mix or penetrate with liquid to prevent the release of particulates. If visible emissions are observed coming from asbestos-containing material, then that material has not been adequately wetted. However, the absence of visible emissions is not sufficient evidence of being adequately wet.

Asbestos: the asbestiform varieties of serpentinite (chrysotile), riebeckite (crocidolite), cummingtonite-grunerite, anthophyllite, and actinolite-tremolite.

Asbestos-containing waste materials: mill tailings or any waste that contains commercial asbestos and is generated by a source subject to the provisions of this subpart. This term includes filters from control devices, friable asbestos waste material, and bags or other similar packaging contaminated with commercial asbestos. As applied to demolition and renovation operations, this term also includes regulated asbestos-containing material waste and materials contaminated with asbestos including disposable equipment and clothing.

Asbestos mill: any facility engaged in converting, or in any intermediate step in converting, asbestos ore into commercial asbestos. Outside storage of asbestos material is not considered a part of the asbestos mill.

Asbestos tailings: any solid waste that contains asbestos and is a product of asbestos mining or milling operations.

Asbestos waste from control devices: any waste material that contains asbestos and is collected by a pollution control device.

Bare soil: Soil not covered with grass, sod, some other similar vegetation, or paving, including the sand in sandboxes.

Building component: Any element of a building that may be painted or have dust on its surface, e.g. walls, stair treads, floors, railings, doors, windowsills, etc.

Category I nonfriable asbestos-containing material (ACM): asbestos containing packings, gaskets, resilient floor covering, and asphalt roofing products containing more than 1 percent asbestos as determined using the method specified in appendix E, subpart E, 40 CFR part 763, section 1, Polarized Light Microscopy.

Category II nonfriable ACM: any material, excluding Category I nonfriable ACM, containing more than 1 percent asbestos as determined using the methods specified in appendix E, subpart E, 40 CFR part 763, section 1, Polarized Light Microscopy that, when dry, cannot be crumbled, pulverized, or reduced to powder by hand pressure.

Certification: The process of testing and evaluating against certain specifications the competence of a person, organization, or other entity in performing a function or service, usually for a specified period of time.

Certified: The designation for Contractors who have completed training and other requirements to safely allow them to undertake risk assessments, inspections, or abatement work. Risk assessors, inspectors, and Abatement Contractors should be certified by the appropriate local, State, or Federal agency.

Chewed surface: Any painted surface that shows evidence of having been chewed or mouthed by a young child. A chewed surface is usually a protruding, horizontal part of a building, such as an interior windowsill.

Cleaning: The process of using a vacuum and wet cleaning agents to remove leaded dust. The process includes the removal of bulk debris from the work area. OSHA prohibits the use of compressed air to clean lead-contaminated dust from a surface.

Clearance examination: Visual examination and collection of environmental samples by an inspector or risk assessor, or, in some circumstances, a Sampling Technician, and analysis by an accredited laboratory upon completion of an abatement project, interim control intervention, or maintenance job that disturbs lead-based paint (or paint suspected of being lead-based). The clearance examination is performed to ensure that lead exposure levels do not exceed standards established by the EPA Administrator pursuant to Title IV of the Toxic Substances Control Act, and that any cleaning following such work adequately meets those standards.

Commercial asbestos: any material containing asbestos that is extracted from ore and has value because of its asbestos content.

Common area: A room or area that is accessible to all residents in a community (e.g. hallways or lobbies). In general, any area not kept locked.

Composite sample: A single sample made up of individual subsamples. Analysis of a composite sample produces the arithmetic mean of all subsamples.

Containment: A process to protect workers and the environment by controlling exposures to the lead-contaminated dust and debris created during abatement.

Cutting: to penetrate with a sharp-edged instrument and includes sawing, but does not include shearing, slicing, or punching.

Demolition: the wrecking or taking out of any load-supporting structural member of a facility together with any related handling operations or the intentional burning of any facility.

Deteriorated lead-based paint: Any lead-based paint coating on a damaged or deteriorated surface or fixture, or any interior or exterior lead-based paint that is peeling, chipping, flaking, worn, chalking, alligatoring, cracking, or otherwise becoming separated from the substrate.

Disposal (of waste): The discharge, deposit, injection, dumping, spilling, leaking, or placement of solid or liquid waste on land or in water so that none of its constituents can pollute the environment by being emitted into the air or discharged into a body of water, including groundwater.

Emergency renovation operation: a renovation operation that was not planned but results from a sudden, unexpected event that, if not immediately attended to, presents a safety or public health hazard, is necessary to protect equipment from damage, or is necessary to avoid imposing an unreasonable financial burden. This term includes operations necessitated by nonroutine failures of equipment.

Encapsulation: Any covering or coating that acts as a barrier between lead-based paint and the environment, the durability of which relies on adhesion and the integrity of the existing bonds between multiple layers of paint and between the paint and the substrate. See also **Enclosure**.

Enclosure: The use of rigid, durable construction materials that are mechanically fastened to the substrate to act as a barrier between the lead-based paint and the environment.

Evaluation: Risk assessment, paint inspection, reevaluation, investigation, clearance examination, or risk assessment screen.

Fabricating: any processing (e.g., cutting, sawing, drilling) of a manufactured product that contains commercial asbestos, with the exception of processing at temporary sites (field fabricating) for the construction or restoration of facilities. In the case of friction products, fabricating includes bonding, debonding, grinding, sawing, drilling, or other similar operations performed as part of fabricating.

Facility: any institutional, commercial, public, industrial, or residential structure, installation, or building (including any structure, installation, or building containing condominiums or individual dwelling units operated as a residential cooperative, but excluding residential buildings having four or fewer dwelling units); any ship; and any active or inactive waste disposal site. For purposes of this definition, any building, structure, or installation that contains a loft used as a dwelling is not considered a residential structure, installation, or building. Any structure, installation or building that was previously subject to this subpart is not excluded, regardless of its current use or function.

Facility component: any part of a facility including equipment.

Federal Register (FR): A daily Federal publication that contains proposed and final regulations, rules, and notices.

Friable asbestos material: any material containing more than 1 percent asbestos as determined using the method specified in appendix E, subpart E, 40 CFR part 763 section 1, Polarized Light Microscopy, that, when dry, can be crumbled, pulverized, or reduced to powder by hand pressure. If the asbestos content is less than 10 percent as determined by a method other than point counting by polarized light microscopy (PLM), verify the asbestos content by point counting using PLM.

Fugitive source: any source of emissions not controlled by an air pollution control device.

Glove bag: a sealed compartment with attached inner gloves used for the handling of asbestos-containing materials. Properly installed and used, glove bags provide a small work area enclosure typically used for small-scale asbestos stripping operations. Information on glove-bag installation, equipment and supplies, and work practices is contained in the Occupational Safety and Health Administration's (OSHA's) final rule on occupational exposure to asbestos (appendix G to 29 CFR 1926.58).

Grinding: to reduce to powder or small fragments and includes mechanical chipping or drilling.

Impact Surface: An interior or exterior surface (such as surfaces on doors) subject to damage by repeated impact or contact.

In poor condition: the binding of the material is losing its integrity as indicated by peeling, cracking, or crumbling of the material.

Inactive waste disposal site: any disposal site or portion of it where additional asbestos-containing waste material has not been deposited within the past year. Installation means any building or structure or any group of buildings or structures at a single demolition or renovation site that are under the control of the same owner or operator (or owner or operator under common control).

Inspection (of paint): A surface-by-surface investigation to determine the presence of lead-based paint (in some cases including dust and soil sampling) and a report of the results.

Interim controls: A set of measures designed to temporarily reduce human exposure or possible exposure to lead-based paint hazards. Such measures include specialized cleaning, repairs, maintenance, painting, temporary containment, and management and resident education programs. Monitoring, conducted by Owners, and reevaluations, conducted by professionals, are integral elements of interim control. Interim controls include dust removal, paint film stabilization, treatment of friction and impact surfaces, installation of soil coverings such as grass or sod, and land use controls. See also **Monitoring, Reevaluations, and Abatement**.

Interior windowsill: The portion of the horizontal window ledge that protrudes into the interior of the room, adjacent to the window sash when the window is closed. Often called the window stool.

Latex: A waterborne emulsion paint made with synthetic binders, such as 100% acrylic, vinyl acrylic, terpolymer, or styrene acrylic. A stable emulsion of polymers and pigment in water.

Lead: Lead includes metallic lead and inorganic and organic compounds of lead.

Lead-based paint: Any paint, varnish, shellac, or other coating that contains lead equal to or greater than 1.0 mg/cm² (milligrams of lead per square centimeter of surface) as measured by XRF or laboratory analysis, or 0.5% by weight (5,000ug/g, 5,000 ppm (parts per million), or 5,000 mg/kg) as measured by laboratory analysis (Local definitions may vary.)

Lead-based paint hazard: A condition in which exposure to lead from lead-contaminated dust, lead-contaminated soil, or deteriorated lead-based paint would have an adverse effect on human health (as established by the EPA Administrator under Title IV of the Toxic Substances Control Act). Lead-based paint hazards include, for example, deteriorated lead-based paint, leaded dust levels above applicable standards. And bare leaded soil above applicable standards.

Lead-based paint hazards control: Activities to control and eliminate lead-based paint hazards, including interim controls, abatement, and complete abatement.

Lead-contaminated dust: Surface dust in residences that contains an area concentration of lead in excess of the standard established by the EPA Administrator, pursuant to Title IV of the Toxic Substances Control Act. As of April 01, 2017, EPA standards for lead dust for risk assessments are ≥ 10 ug/ft² (micrograms of lead per square foot) for floors, and ≥ 100 ug/ft² for interior windowsills. The EPA standard for clearance are < 10 ug/ft² for floors, 100 ug/ft² for windowsills, and 100 ug/ft² for window troughs. Porch floors are also wiped during final clearance, with a lead dust clearance action level of < 40 ug/ft².

Lead-contaminated soil: Bare soil on residential property that contains lead in excess of the standard established by the EPA Administrator, pursuant to Title IV of the Toxic Substances Control Act. The standard is 400 ug/g in play areas and 1,200 ug/g in the rest of the property.

Leak-tight: solids or liquids cannot escape or spill out. It also means dust-tight.

Licensed: Holding a valid license or certification issued by the EPA or by an EPA-approved State program pursuant to Title IV of the Toxic Substances Control Act. The license is based on certification for lead-based paint hazard control work. See also **Certified**.

Maintenance: Work intended to maintain adequate living conditions in a dwelling, which has the potential to disturb lead-based paint or paint that is suspected of being lead-based.

Malfunction: any sudden and unavoidable failure of air pollution control equipment or process equipment or of a process to operate in a normal or usual manner so that emissions of asbestos are increased. Failures of equipment shall not be considered malfunctions if they are caused in any way by poor maintenance, careless operation, or any other preventable upset conditions, equipment breakdown, or process failure.

Manufacturing: the combining of commercial asbestos-or, in the case of woven friction products, the combining of textiles containing commercial asbestos-with any other material(s), including commercial asbestos, and the processing of this combination into a product. Chlorine production is considered a part of manufacturing.

Mean: The arithmetic average of a series of numerical data values. For example, the algebraic sum of the data values divided by the number of data values.

Microgram (ug): 1/1,000,000 of a gram. Used to measure weight

Monitoring: Surveillance to determine (1) that known or suspected lead-based paint is not deteriorating, (2) that lead-based paint hazard controls, such as paint stabilization, enclosure, or encapsulation have not failed, and (3) that structural problems do not threaten the integrity of hazard controls.

Natural barrier: a natural object that effectively precludes or deters access. Natural barriers include physical obstacles such as cliffs, lakes or other large bodies of water, deep and wide ravines, and mountains. Remoteness by itself is not a natural barrier.

Nonfriable asbestos-containing material: any material containing more than 1 percent asbestos as determined using the method specified in appendix E, subpart E, 40 CFR part 763, section 1, Polarized Light Microscopy, that, when dry, cannot be crumbled, pulverized, or reduced to powder by hand pressure.

Nonscheduled renovation operation: a renovation operation necessitated by the routine failure of equipment, which is expected to occur within a given period based on past operating experience, but for which an exact date cannot be predicted.

Outside air: the air outside buildings and structures, including, but not limited to, the air under a bridge or in an open-air ferry dock.

Owner: A person, firm, corporation, guardian, conservator, receiver, trustee, executor, government agency or entity, or other judicial officer who, alone or with others, owns, holds, or controls the freehold or leasehold title or part of the title to property, with or without actually possessing it. This definition includes a vendee who possesses the title, but does not include a mortgagee or an Owner of a reversionary interest under a ground rent lease.

Paint inspector: An individual who has completed training from an accredited program and been licensed or certified by the appropriate State or local agency to (1) perform inspections to determine and report the presence of lead-based paint on a surface-by-surface basis through onsite testing, (2) report the findings of such an inspection, (3) collect environmental samples for laboratory analysis, (4) perform clearance testing, and optionally (5) document successful compliance with lead-based paint hazard control requirements or standards.

Paint removal: An abatement strategy that entails the removal of lead-based paint from surfaces. For lead hazard control work, this can mean using chemicals, heat guns below 1,100°F, and certain contained abrasive methods. Open-flame burning, open-abrasive blasting, sandblasting, extensive dry scraping, and stripping in a poorly ventilated space using a volatile stripper are prohibited paint removal methods. Hydroblasting is not recommended.

Particulate asbestos material: finely divided particles of asbestos or material containing asbestos.

Planned renovation operations: a renovation operation, or a number of such operations, in which some RACM will be removed or stripped within a given period of time and that can be predicted. Individual nonscheduled operations are included if a number of such operations can be predicted to occur during a given period of time based on operating experience.

Polyethylene plastic: All references to polyethylene plastic refer to 6 mil plastic sheeting or polyethylene bags (or double bags if using 4 mil polyethylene bags), or any other thick plastic material shown to demonstrate at least the equivalent dust contamination performance. Plastic used to contain waste should be capable of completely containing the waste and, after being properly sealed, should remain leak tight with no visible signs of discharge during movement or relocation.

Polyurethane: An exceptionally hard and wear-resistant coating (created by the reaction of polyols with a multifunctional isocyanate). Often used to seal wood floors following lead-based paint hazard control work and cleaning.

Reevaluation: In lead hazard control work, the combination of a visual assessment and collection of environmental samples performed by a certified risk assessor to determine if a previously implemented lead-based paint hazard control measure is still effective and if the dwelling remains lead-safe.

Regulated asbestos-containing material (RACM): (a) Friable asbestos material, (b) Category I nonfriable ACM that has become friable, (c) Category I nonfriable ACM that will be or has been subjected to sanding, grinding, cutting, or abrading, or (d) Category II nonfriable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of demolition or renovation operations regulated by this subpart.

Remove: to take out RACM or facility components that contain or are covered with RACM from any facility.

Renovation: altering a facility or one or more facility components in any way, including the stripping or removal of RACM from a facility component. Operations in which load-supporting structural members are wrecked or taken out are demolitions.

Replacement: A strategy of abatement that entails the removal of building components coated with lead-based paint (such as windows, doors, and trim) and the installation of new components free of lead-based paint.

Resident: A person who lives in a dwelling.

Resilient floor covering: asbestos-containing floor tile, including asphalt and vinyl floor tile, and sheet vinyl floor covering containing more than 1 percent asbestos as determined using polarized light microscopy according to the method specified in appendix E, subpart E, 40 CFR part 763, Section 1, Polarized Light Microscopy.

Risk assessment: An onsite investigation of a residential dwelling to discover any lead-based paint hazards. Risk assessments include an investigation of the age, history, management, and maintenance of the dwelling, and the number of children under age 6 and women of childbearing age who are residents; a visual assessment; limited environmental sampling (i.e. collection of dust wipe samples, soil samples, and deteriorated paint samples); and preparation of a report identifying acceptable abatement and interim control strategies based on specific conditions.

Risk assessor: A certified individual who has completed training with an accredited training program and who has been certified to (1) perform risk assessments, (2) identify acceptable abatement and interim control strategies for reducing identified lead-based paint hazards, (3) perform clearance testing and reevaluations, and (4) document the successful completion of lead-based paint hazard control activities.

Roadways: surfaces on which vehicles travel. This term includes public and private highways, roads, streets, parking areas, and driveways.

Site: The land or body of water where a facility is located or an activity is conducted. The site includes adjacent land used in connection with the facility or activity.

Spectrum analyzer: A type of XRF analyzer that provides the operator with a plot of the energy and intensity, or counts of both K and L x-ray spectra, as well as a calculated lead concentration. See also **XRF analyzer**.

Standard deviation: A measure of the precision of a reading. The spread of the deviation from the mean. The smaller the standard deviation, the more precise the analysis. The standard deviation is calculated by first obtaining the mean, or the arithmetic average, of all of the readings. A formula is then used to calculate how much the individual values vary from the mean. the standard deviation is the square root of the arithmetic average of the squares of the deviation from the mean. Many hand calculators have an automatic standard deviation function.

Strip: to take off RACM from any part of a facility or facility components.

Structural member: any load-supporting member of a facility, such as beams and load supporting walls; or any non-load-supporting member, such as ceilings and non-load-supporting walls.

Subsample: A representative portion of a sample. A subsample may be either a field sample or a laboratory sample. A subsample is often combined with other subsamples to produce a composite sample. See also **Composite sample**.

Substrate: A surface on which paint, varnish, or other coating has been applied or may be applied. Examples of substrates include wood, plaster, metal, and drywall.

Substrate effect: The radiation returned to an XRF analyzer by the paint, substrate, or underlying material, in addition to the radiation returned by any lead present. This radiation, when counted as lead x-rays by an XRF analyzer contributes to substrate equivalent lead (bias). The inspector may have to compensate for this effect when using XRF analyzers. See also **XRF analyzer**.

Substrate Equivalent Lead (SEL): The XRF measurement taken on an unpainted surface, used to calculate the corrected lead concentration on a surface by using the following formula: Apparent Lead Concentration-Substrate Equivalent Lead = Corrected Lead Concentration. See also, **XRF analyzer**.

Target housing: Any residential unit constructed before 1978, except dwellings that do not contain bedrooms or dwellings that were developed specifically for the elderly or persons with disabilities, unless a child younger than 6 resides or is expected to reside in the dwelling. In the case of jurisdictions that banned the sale or use of lead-based paint before 1978, the Secretary of HUD may designate an earlier date for defining target housing.

Test location: A specific area on a testing combination where XRF instruments will test for lead-based paint.

Trained: Successful completion of a training course in a particular discipline. For lead hazards control work, the training course must be accredited by the EPA or by an EPA-approved State program, pursuant to Title IV of the Toxic Substances Control Act.

Treatment: In residential lead-based paint hazard control work, any method designed to control lead-based paint hazards. Treatment includes interim controls, abatement, and removal.

Trough: See **Window trough**.

Visible emissions: any emissions, which are visually detectable without the aid of instruments, coming from RACM or asbestos-containing waste material, or from any asbestos milling, manufacturing, or fabricating operation. This does not include condensed, uncombined water vapor.

Waste generator: any owner or operator of a source covered by this subpart whose act or process produces asbestos-containing waste material.

Waste shipment record: the shipping document, required to be originated and signed by the waste generator, used to track and substantiate the disposition of asbestos-containing waste material.

Windowsill: See **Interior windowsill**.

Window trough: For a typical double-hung window, the portion of the exterior windowsill between the interior windowsill (or stool) and the frame of the storm window. If there is no storm window, the window trough is the area that receives both the upper and lower window sashes when they are both lowered. Sometimes inaccurately called the window well.

Worker: An individual who has completed training in an accredited program to perform lead-based paint hazard control in housing.

Working day: Monday through Friday and includes holidays that fall on any of the days Monday through Friday.

Worksite: Any interior or exterior area where lead-based paint hazard control work takes place.

XRF analyzer: An instrument that determines lead concentration in milligrams per square centimeter (mg/cm²) using the principle of x-ray fluorescence (XRF). Two types of field portable XRF analyzers are used. direct readers and spectrum analyzers. For this lead-based paint inspection, the term XRF analyzer only refers to portable instruments manufactured to analyze paint, that have a HUD Performance Characteristic Sheet, and are interpreted in accordance with the Performance Characteristic Sheet. It does not refer here to laboratory-grade units or portable instruments designed to analyze soil.

NESHAP (40CFR Parts 61 and 63, Subpart M)- regulates the removal, demolition, and/or renovation of buildings and structures with ACM. Also provides specific standards for inactive asbestos waste disposal facilities, and requirements for covering asbestos-containing waste materials, engineering controls to limit site access and signage.